

GCI Basic Learning Materials

Course Overview

This course covers a wide range of topics from data science fundamentals to Python basic syntax, as well as introductions to statistics and machine learning.

It is specifically designed from a beginner's perspective, providing comprehensive knowledge to help those with no prior background smoothly continue their GCI studies.

By completing this entire course, you will be able to create basic predictive models (programs that perform future predictions using data from your surroundings).

The estimated study time is 6 hours. Each video is designed to be under 5 minutes long, so please make use of commuting time and other spare moments to advance your learning.

After GCI begins, Python basic syntax will be covered in the 2nd lecture "Basics of Python" so it is recommended to complete at least "2. Basics of Python" before the 2nd lecture.

Materials Storage Location

How to Use GCI Basic

<https://www.youtube.com/playlist?list=PLiEF5y87wtyySsLMrL4eRbvVWPm97ST04>

0. Opening

<https://youtu.be/q2tRzyQ27A>

1. What is Data Science?

Deepen understanding of the overall picture and basic flow of data science.

<https://www.youtube.com/playlist?list=PLiEF5y87wtyxPTjPp7hrbg6mmOmdWOFj->

2. Basics of Python

Learn the characteristics and basic operations of Python, a programming language frequently used in data science.

https://www.youtube.com/playlist?list=PLiEF5y87wtyyc3mVmmRpUcAi_a74w52Cl

3. Basics of Statistics

Acquire the necessary ways of thinking to understand the characteristics and meaning of the data you handle.

<https://www.youtube.com/playlist?list=PLiEF5y87wtyscNTwmUw4OtUwOGFGUrgNq>

4. What is Machine Learning?

Understand machine learning concepts and representative methods.

<https://www.youtube.com/playlist?list=PLiEF5y87wtysJUa9JJZWs30sAU40-BwC1>

5. Review

Organize the content covered so far and confirm comprehensive understanding.

<https://youtu.be/e1nyWJ7eXml>

6A. Exercise: Regression

Develop the ability to actually program on your own

<https://www.youtube.com/playlist?list=PLiEF5y87wtys7MepWS41WqR-tTF8FqzY8>

6B. Exercise: Classification

Develop the ability to actually program on your own

<https://www.youtube.com/playlist?list=PLiEF5y87wtysaOJK9AvxIKp0YgX246IFr>

- Learning Materials Storage Location ※ Same location as this document

https://drive.google.com/drive/folders/1M_lr5GCq7aIS6Wt6Ei3TiXG8LQ-Ndwmv

How to Use the Materials

- Watch the videos and perform exercises on the notebooks to solidify your knowledge.

It may be a good idea to prepare two separate monitors so you can have the videos and notebooks on different screens for more efficient learning.

- Ask questions on Slack's Q&A channel or during OH (Office Hours)